

02

Respiratory Care

Differentiate respiratory distress from respiratory failure and initiate safe airway support, including non-invasive respiratory support before intubation.

2. Respiratory Care

Learning Objectives

- Classify a respiratory presentation as upper airway obstruction, lower airway obstruction, lung tissue disease, or disordered control of breathing.
- Select the correct oxygen delivery device using the AIM framework (Assess, Identify, Match).
- Perform safe two-person bag-mask ventilation and initiate HFNC.
- Deliver a high-quality nebulisation and titrate bronchodilator therapy for acute wheeze.
- Recognise when non-invasive support is failing and escalation to invasive ventilation is required.

1. Distress vs Failure — Classify the Pattern



Upper Airway Obstruction

Foreign body, croup, epiglottitis. Signs: stridor, voice change, drooling/gurgling, poor chest rise.



Lower Airway Obstruction

Asthma, bronchiolitis. Signs: ↑RR & effort, wheeze, cough, ↓air movement.



Lung Tissue Disease

Pneumonia, oedema, ARDS. Signs: tachypnoea, grunting, crackles, hypoxaemia.



Disordered Control

CNS infection, TBI, sedative overdose. Signs: irregular rate, shallow breaths, apnoea.



ESCALATE IMMEDIATELY

Airway procedures generate aerosol — use PPE. A child with risk factors needs close monitoring or early referral before distress becomes failure.

2. First Steps — Open & Support the Airway

1

Open the airway

Correct alignment; head-tilt–chin-lift, or jaw-thrust if spine injury suspected



2

Adjunct if needed

Oropharyngeal if unresponsive/no gag; nasopharyngeal if semi-conscious (avoid in facial/skull injury)



3

Tracheostomy in situ?

Suction, or change tube if blockage not clearing with suction



4

Check SpO₂

<94% on air → supplemental O₂, titrate to 95–98% (100% only in shock via NRM)

3. Oxygen Devices — the AIM Framework

A
Assess

Patient need

I
Identify

Technology available

M
Match

Technology to need

Device	Flow	FiO ₂	Notes
Nasal prongs	≤4 L/min	Low, variable	Mild hypoxia
Simple face mask	6–10 L/min	Variable	Flow < minute ventilation
Non-rebreathing mask	6–10 L/min	>95%	Sick child / shock
HFNC	Up to 2 L/kg/min	0.21–1.0	Some PEEP effect, dead-space washout

Breathing still inadequate despite O₂? Support with positive-pressure bag-mask breaths — a core peri-arrest skill.

4. Bag-Mask Ventilation (BMV)


Good Seal

Age-appropriate mask; allow full inspiratory time to avoid breath-stacking


2-Person Technique

One holds E-C clamp seal, one squeezes the bag — use when seal is hard alone


Escalate

LMA if competent; intubation only per RSI principles by a trained provider

5. Non-Invasive Ventilation & HHHFNC

Type	Definition	Common causes
I — Hypoxaemic	PaO ₂ <60, normal/low CO ₂	Pneumonia, mild ARDS, oedema, bronchiolitis
II — Hypercapnic	PaCO ₂ >50, hypoxaemia common	Impaired drive, neuromuscular weakness, chest wall issue


WHEN NIV IS FAILING

Worsening work of breathing, rising CO₂, or falling consciousness → escalate promptly to invasive ventilation (Section 12). Also useful in shock/cardiac failure to offload the work of breathing.

6. Pneumonia — Quick Bundle



Monitor

Continuous cardio-resp + SpO₂



Support

Position, clear airway, high-flow O₂,
bronchodilator if needed



Treat

Antipyretics; antibiotics within 1st
hour; consider fluids

7. Acute Wheeze — Drug Doses



NEBULISED SALBUTAMOL

0.15 mg/kg (min 2.5, max 5 mg) every 20 min ×3, then continuous 0.5 mg/kg/hr if needed. ±
Budesonide/Ipratropium (250 mcg <20 kg; 500 mcg >20 kg). O₂-driven at 6–8 L/min —
recheck SpO₂, RR, distress every few minutes.

Adjunct	Dose
Prednisolone	2 mg/kg PO, then 1 mg/kg OD
Dexamethasone	0.3–0.6 mg/kg (max 12–16 mg/day)
Methylprednisolone	1–2 mg/kg/day IV
Magnesium sulfate	50–100 mg/kg over 0.5–1 hr



LINKS TO THE HANDS-ON LAB

Oxygen-device selection stations, HFNC setup, nebuliser technique, and two-person BMV
practice — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Section 3: Basic Management & O₂ Delivery Devices; Section 4: Management of Respiratory Emergencies (pp. 15–30).
- BMV Workshop Manual (PediCritCon 2025) — Non-Invasive Ventilation & HHHFNC (pp. 6–18).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.